Study Guide: The Molecular Basis of Inheritance

This study guide provides a comprehensive review of the molecular mechanisms underlying genetics, including the structure and function of DNA and RNA, the processes of replication, transcription, and translation, and the regulation of gene expression.

🧬 Molecular Basis of Inheritance: Comprehensive Notes

1. Structure of Polynucleotide Chain (DNA & RNA)

DNA (Deoxyribonucleic acid) is a long polymer of deoxyribonucleotides, and its length is defined by the number of base pairs. A nucleotide has three components: a **nitrogenous base**, a **pentose sugar**, and a **phosphate group**. Purines include **Adenine (A)** and **Guanine (G)**, while Pyrimidines include **Cytosine (C)**, **Thymine (T)** (in DNA), and **Uracil (U)** (in RNA).

- **Watson-Crick Double Helix Model:** DNA consists of two polynucleotide chains with a sugar-phosphate backbone. The chains are **antiparallel** (one has 5'→3' polarity, the other 3'→5') and coil in a right-handed fashion. Bases are paired via hydrogen bonds: **A pairs with T** (2 H-bonds) and **G pairs with C** (3 H-bonds). The pitch of the helix is 3.4 nm with roughly 10 base pairs per turn.
- **Chargaff's Rule:** The ratios between Adenine and Thymine, and Guanine and Cytosine are constant and equal to one.

2. Packaging of DNA Helix

In prokaryotes, the negatively charged DNA is held together with positive proteins in a region called the **nucleoid**. In eukaryotes, DNA wraps around a positively charged **histone octamer** (rich in basic amino acids lysine and arginine) to form a **nucleosome**.

Types of Chromatin:

- **Euchromatin**: Loosely packed, stains light, and is transcriptionally active.
- **Heterochromatin**: Densely packed, stains dark, and is transcriptionally inactive.

3. The Search for Genetic Material

- **Griffith's Experiment (1928)**: He infected mice with *Streptococcus pneumoniae*. He found that a "transforming principle" from heat-killed S-strain (virulent) bacteria could transform live R-strain (non-virulent) bacteria into a virulent form.

- **Avery, MacLeod, & McCarty**: They discovered that only **DNase** (DNA-digesting enzyme) inhibited transformation, proving DNA is the hereditary material.
- **Hershey & Chase Experiment (1952)**: They provided unequivocal proof using bacteriophages. They labeled viral DNA with radioactive phosphorus (32P) and viral proteins with radioactive sulfur (35S). Only the radioactive DNA entered the *E. coli* bacteria, confirming DNA is the genetic material.

🦠 Hershey & Chase Experiment Flowchart:
Bacteriophage + E. coli
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Infection (Viral genetic material enters bacteria)
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Blending (Viral coats removed from bacterial surface)
↓
Centrifugation (Separation of heavier bacteria from lighter viral coats)

4. DNA Replication

DNA replication is **semiconservative**, meaning each newly synthesized DNA molecule contains one parental strand and one new strand. This was experimentally proven by **Meselson and Stahl** using heavy nitrogen (15N) isotopes.

Mechanism & Enzymes Flowchart:

1. **Unwinding**: The enzyme **Helicase** opens the DNA double helix to form a replication fork.
2. **Stabilization**: **Single Stranded Binding Proteins (SSBPs)** keep the separated strands from re-coiling.
3. **Tension Release**: **Topoisomerase** releases the tension created by unwinding.
4. **Initiation**: **Primase** synthesizes a short RNA primer.
5. **Elongation**: **DNA-dependent DNA Polymerase III** continuously adds nucleotides in the 5'→3' direction on the **leading strand**.
6. **Discontinuous Synthesis**: On the opposite strand, DNA is synthesized in short stretches called **Okazaki fragments** (the **lagging strand**).
7. **Joining**: **DNA Ligase** joins the Okazaki fragments together.

5. Transcription

Transcription is the process of copying genetic information from one strand of DNA into RNA. A **transcription unit** contains a Promoter, a Structural gene, and a Terminator.

- **In Prokaryotes**: A single DNA-dependent RNA polymerase catalyzes the transcription of all types of RNA. The σ (sigma) factor initiates the process, and the ρ (rho) factor terminates it.
- **In Eukaryotes**: There are three RNA polymerases (I for rRNA, II for mRNA/hnRNA, and III for tRNA). The primary transcript (hnRNA) must undergo post-transcriptional modifications to become mature mRNA.

✂️ Eukaryotic RNA Modification Flowchart:
Splicing (Removal of non-coding introns & joining of coding exons)
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Capping (Addition of methyl guanosine triphosphate at the 5' end)
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Tailing (Addition of 200-300 adenylate residues at the 3' end)

6. Genetic Code & Mutations

The genetic code dictates the sequence of amino acids during protein synthesis.

- It is a **triplet code** (61 codons code for amino acids, and 3 are stop codons: UAA, UAG, UGA).
- It is **unambiguous** (one codon specifies one amino acid) and **degenerate** (some amino acids are coded by more than one codon).
- The code is read continuously with no punctuations, and is nearly **universal** across all organisms.
- **AUG** has a dual function: it codes for Methionine and serves as the initiator codon.
- Mutations like insertions or deletions can cause **frameshift mutations**, altering the reading frame from the point of change.

7. Translation (Protein Synthesis)

Translation is the process of polymerizing amino acids to form a polypeptide chain. **Ribosomes** act as the cellular protein factories.

Translation Flowchart:

1. **Charging of tRNA**: Amino acids are activated with ATP and linked to their specific **tRNA** (adapter molecule).
2. **Initiation**: The small ribosomal subunit binds to mRNA at the AUG start codon, and the initiator tRNA attaches.
3. **Elongation**: New charged tRNAs enter the ribosome, and peptide bonds are formed between amino acids.
4. **Termination**: A stop codon is reached, and a release factor frees the completed polypeptide chain.

8. Regulation of Gene Expression (Lac Operon)

Gene expression can be regulated depending on cellular requirements. An **operon** is a polycistronic structural gene regulated by a common promoter.

- **Components of Lac Operon**: The **i gene** codes for the repressor, the **z gene** codes for β-galactosidase, the **y gene** codes for permease, and the **a gene** codes for transacetylase.
- **Switch OFF**: In the absence of lactose, the repressor binds to the operator, blocking RNA polymerase from transcribing the genes.
- **Switch ON**: When lactose (**the inducer**) is present, it binds to the repressor and inactivates it, allowing transcription to proceed.

9. Human Genome Project (HGP)

The HGP was a mega project aimed at sequencing the entire human genome.

- The human genome contains 3,164.7 million base pairs, and humans have an estimated 30,000 genes.
- Less than 2% of the genome codes for proteins.
- Chromosome 1 has the most genes (2968), and the Y chromosome has the fewest (231).

10. DNA Fingerprinting

Developed by Alec Jeffreys, this technique identifies differences in specific DNA regions called **Variable Number of Tandem Repeats (VNTRs)**

Isolation of DNA

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Digestion by restriction endonucleases
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Separation of fragments by gel electrophoresis
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Southern blotting onto synthetic membranes
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Hybridization with radiolabeled VNTR probes
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Detection by autoradiography

Part 1: Short Answer Quiz

Instructions: Answer the following ten questions in 2–3 sentences.

1. **What is the "Central Dogma of Molecular Biology," and who proposed it?**
2. **How did Frederick Griffith's experiment with *Streptococcus pneumoniae* demonstrate the existence of a "transforming principle"?**
3. **Explain the specific roles of the radioactive isotopes ^{32}P and ^{35}S in the Hershey-Chase experiment.**
4. **Describe the structural components and charge of a nucleosome.**
5. **Why does DNA replication occur discontinuously on one of the parental strands?**
6. **Contrast the chemical stability of DNA and RNA based on their molecular structures.**
7. **What are the dual functions of the AUG codon during protein synthesis?**
8. **Briefly explain the process of RNA splicing in eukaryotic cells.**
9. **In the context of the *lac* operon, what is the role of the inducer?**
10. **What is DNA polymorphism, and why is it central to the technique of DNA fingerprinting?**

Part 2: Answer Key

1. **The Central Dogma:** Proposed by Francis Crick, the Central Dogma describes the unidirectional flow of genetic information from DNA to DNA (replication), DNA to mRNA (transcription), and mRNA to proteins (translation). It serves as the fundamental framework for understanding how genetic instructions are expressed in biological systems.
2. **Griffith's Transforming Principle:** Griffith observed that when heat-killed S strain (virulent) bacteria were mixed with live R strain (non-virulent) bacteria and injected into mice, the mice died. He concluded that a "transforming principle" had transferred from the dead S strain to the live R strain, enabling the latter to synthesize a protective polysaccharide coat and become virulent.
3. **Hershey-Chase Isotopes:** Radioactive phosphorus (^{32}P) was used to label viral DNA because DNA contains phosphorus but protein does not. Radioactive sulfur (^{35}S) was used to label the viral protein coat because sulfur is present in certain amino acids (cysteine and methionine) but absent in DNA; this allowed the researchers to track which molecule entered the bacteria during infection.
4. **Nucleosome Structure:** A nucleosome consists of approximately 200 base pairs of negatively charged DNA wrapped around a positively charged octamer of histone proteins (two each of H2A, H2B, H3, and H4). This structure is the basic unit of chromatin packaging in eukaryotes, appearing as "beads-on-a-string" under an electron microscope.
5. **Discontinuous Synthesis:** DNA polymerase can only synthesize new strands in the 5'→3' direction. Because the two strands of the DNA double helix are antiparallel, only one strand (the leading strand) can be synthesized continuously toward the replication fork, while the other (the lagging strand) must be synthesized in short, discontinuous stretches called Okazaki fragments.
6. **DNA vs. RNA Stability:** DNA is more stable because it lacks the reactive 2'-OH group found in ribose sugar and utilizes thymine instead of uracil, which confers additional resistance to degradation. RNA is highly reactive and labile due to the presence of the

2'-OH group at every nucleotide, making it more prone to mutations and enzymatic breakdown.

7. **Functions of AUG:** The AUG codon serves a dual purpose: it acts as the initiator codon (start codon) that signals the beginning of translation, and it codes for the amino acid Methionine (met). This ensures that the protein synthesis machinery starts at the correct position on the mRNA strand.
8. **RNA Splicing:** RNA splicing is a post-transcriptional modification in eukaryotes where non-coding regions called introns are removed from the primary transcript (hnRNA). The remaining coding regions, known as exons, are joined together to form a continuous, functional mRNA sequence that can be translated into a protein.
9. **The Inducer in Lac Operon:** In the *lac* operon, the inducer (lactose) binds to the repressor protein, causing a conformational change that prevents the repressor from binding to the operator gene. This allows RNA polymerase to access the promoter and initiate the transcription of structural genes (lacZ, lacY, and lacA) needed for lactose metabolism.
10. **DNA Polymorphism:** DNA polymorphism refers to variations in the DNA sequence, often occurring in non-coding regions, that arise due to mutations. These variations create unique, highly variable patterns (such as VNTRs) in an individual's genome, which can be separated and detected via DNA fingerprinting for identification purposes.

Part 3: Essay Questions

Instructions: These questions are designed for in-depth study and require the synthesis of multiple concepts from the source material.

1. **The Evolution of Genetic Evidence:** Trace the historical progression of experiments that identified DNA as the genetic material. Compare the methodology and persuasiveness of Griffith's transformation experiment, Avery-MacLeod-McCarty's biochemical characterization, and the Hershey-Chase bacteriophage study.
2. **Mechanisms of Fidelity in Replication:** Detail the process of DNA replication in *E. coli* , emphasizing the roles of DNA polymerase I and III, helicase, primase, and ligase. Discuss how the cell ensures accuracy through proofreading and the energetic requirements of this process.
3. **Complexity of Eukaryotic Gene Expression:** Describe the additional steps required for gene expression in eukaryotes compared to prokaryotes. Focus on mRNA processing (capping, tailing, and splicing) and the different levels of regulation, from the transcriptional to the translational stage.
4. **The Nature of the Genetic Code:** Discuss the salient features of the genetic code, including its triplet nature, universality, degeneracy, and lack of punctuation. Explain how George Gamow and other scientists deciphered the relationship between four nitrogenous bases and twenty amino acids.
5. **Genomic Research and the Human Genome Project:** Analyze the key findings of the Human Genome Project. Discuss the significance of repetitive DNA sequences and the future challenges scientists face in deriving meaningful biological knowledge from whole-genome sequencing.

Part 4: Glossary of Key Terms

Term,Definition
Antiparallel,"The orientation of the two strands in a DNA double helix, where one strand runs in the 5'→3' direction and the other in the 3'→5' direction."
Bacteriophage,A type of virus that infects and replicates within bacteria; used by Hershey and Chase to prove DNA is genetic material.
Cistron,"A segment of DNA coding for a single polypeptide chain, including structural and regulatory sequences."
Codon,A sequence of three nucleotides on mRNA that corresponds to a specific amino acid or a stop signal during protein synthesis.
DNA Fingerprinting,A technique used to identify individuals by analyzing unique patterns of repetitive DNA sequences (satellite DNA).
DNA Polymerase,"An enzyme that catalyzes the polymerization of deoxyribonucleotides into a DNA strand, typically in the 5'→3' direction."
Euchromatin,"Loosely packed, lightly stained regions of chromatin that are transcriptionally active."
Exons,"The coding or expressed sequences in a eukaryotic gene that appear in the mature, processed RNA."
Heterochromatin,"Densely packed, darkly stained regions of chromatin that are transcriptionally inactive and late-replicating."
Histones,Positively charged basic proteins rich in lysine and arginine that facilitate DNA packaging in eukaryotes.
Introns,Non-coding intervening sequences in a eukaryotic gene that are removed during RNA splicing.
Leading Strand,The DNA strand that is synthesized continuously in the 5'→3' direction toward the replication fork.
Nucleoside,A molecule formed by a nitrogenous base linked to a pentose sugar via an N-glycosidic linkage.
Nucleotide,"The monomeric unit of nucleic acids, consisting of a nitrogenous base, a pentose sugar, and a phosphate group."
Okazaki Fragments,"Short, discontinuously synthesized segments of DNA produced on the lagging strand during replication."
Operon,"A functional unit of DNA containing a cluster of genes controlled by a single promoter and operator, common in prokaryotes."
Poly-A Tail,A sequence of 200–300 adenylate residues added to the 3' end of hnRNA to provide longevity and protection from degradation.
Promoter,A specific DNA sequence where RNA polymerase binds to initiate the process of transcription.
Purines,Nitrogenous bases with a double-ring structure; specifically Adenine (A) and Guanine (G).
Pyrimidines,"Nitrogenous bases with a single-ring structure; specifically Cytosine (C), Thymine (T), and Uracil (U)."
Replication Fork,The Y-shaped opening formed when the two strands of a DNA double helix separate during replication.

Semiconservative,The mode of DNA replication where each daughter DNA molecule retains one original parental strand and one newly synthesized strand.
Transcription,The process of copying genetic information from a DNA template strand into a complementary RNA strand.
Translation,The process by which the sequence of codons in mRNA is decoded by ribosomes to synthesize a polypeptide chain.
VNTRs,Variable Number of Tandem Repeats; short DNA sequences repeated multiple times that vary in number between individuals.